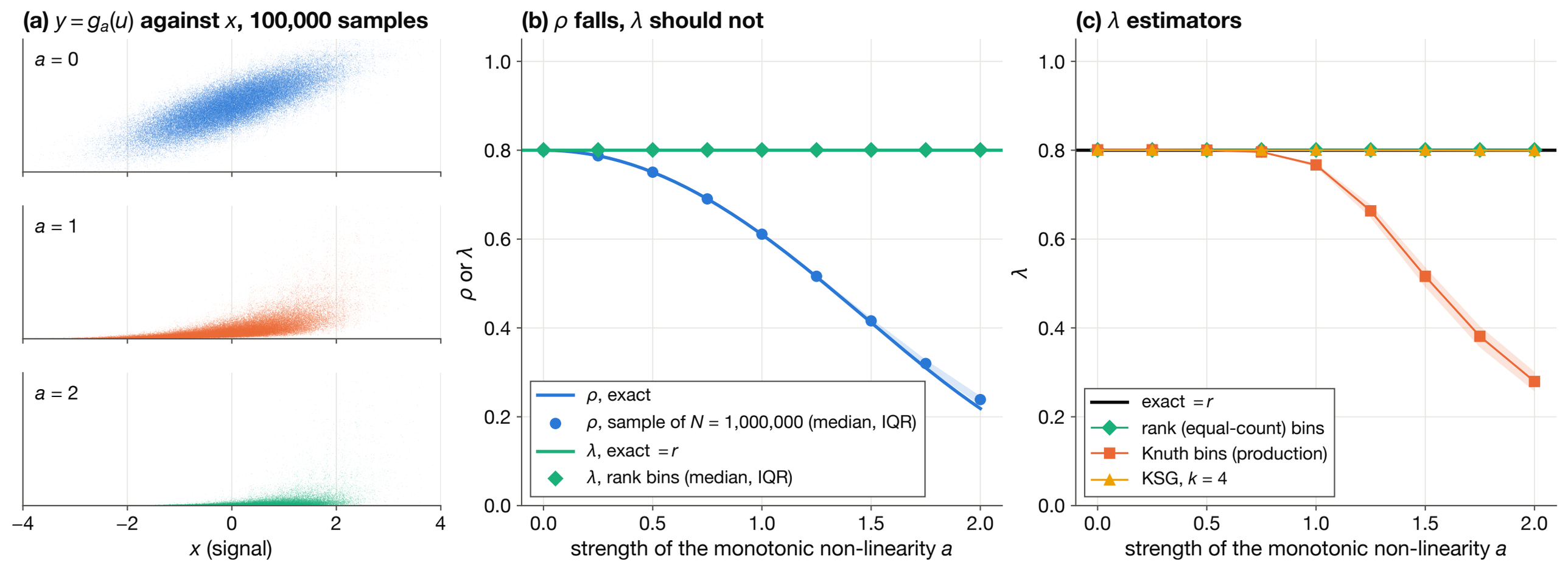

(a) $y = g_a(u)$ against x, 100,000 samples
a = 0
a = 1
a = 2
−4
−2
0
2
4
x (signal)
(b) ρ falls, λ should not
1.0
0.8
0.6
0.4
0.2
0.0
ρ or λ
0.0
0.5
1.0
1.5
2.0
strength of the monotonic non-linearity a
ρ, exact
ρ, sample of N = 1,000,000 (median, IQR)
λ, exact $= r$
λ, rank bins (median, IQR)
(c) λ estimators
λ
exact $= r$
rank (equal-count) bins
Knuth bins (production)
KSG, k = 4